Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/01/2009
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cavanagh@u.washington.edu	**Fax:**	FY 206-685-3139
PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2005	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Meck, J@n	**Contact Phone:**	281-244-5405
Contact Email:	janice.v.meck@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: PI moved to University of Washington in summer 2008 and end date changed to 9/30/2008 (from 5/31/2009); new grant will be issued for remainder of this award, per J. Dardano/JSC (8/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Brian (Cleveland Clinic) Licata, Angelo (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNJ05HE72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involve quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on DLS theory. Subjects were randomized to control or exercise groups and confined to 6-degree head-down bedrest in the General Clinical Research Center (GCRC) at the Cleveland Clinic for 12 weeks. During this time, the exercise group underwent individualized daily exercise programs in the unique Zero Gravity Locomotion Simulator (ZLS) at CCF designed to replace their daily mechanical load stimulus experienced during free living. DXA, MRI, and QCT scans were conducted at the start and end of bedrest to assess variations in bone mineral density, changes in muscle volume, and regional changes (cortical vs. trabecular) in bone. Strength measurements were collected pre- and post-bedrest, and urinary and serum markers of bone resorption and formation were also assessed. Dietary intake sufficient to maintain constant body weight with a balanced intake of macro- and micronutrients was planned and supervised by a registered dietician, and a physician was available 24 hr/day to attend to any medical needs. A rehabilitation program was offered to subjects in the 8 weeks following bedrest. The proposed experiment should provide a categorical answer to the question of whether intermittent load replacement can adequately protect the musculoskeletal system against hypokinetic osteopenia and muscle atrophy and may open a new era in individualized exercise countermeasure planning. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in BMD, bone quality, and muscle atrophy during 12 weeks of bedrest. The goals of the proposed study are congruent with the goals identified in the critical path roadmap for countermeasure development. We will address the following Enabling Question 1e of the Bioastronautics Roadmap: What are the specifics of the optimal exercise regimen with regard to mode, duration, intensity and frequency, to be followed during exposure to hypogravity so as to minimize decreases in bone mass? Is impact loading an essential element and, if so, how can it be produced in hypogravity?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The results from the subjects completed at the Cleveland Clinic are encouraging in that they demonstrate that exercise appears to be effective in not only preserving bone, but also in the increasing bone formation. It was surprising to see the BMD gain in the lumbar spine in both groups but this observation is consistent with anecdotal findings from other bedrest experiments. It was anticipated that there would be attrition of two subjects in the entire study period and thus the study is still adequately powered. During the project period to date at the Cleveland Clinic a number of other milestones have been accomplished. In collaboration with an NSBRI-funded project, the NASA Glenn Research Center and CCF project team developed a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. Our research group has also developed a customized harnesses with bilayer-padded shoulder straps. The project team has also modified existing mathematical theories on daily load stimulus (DLS) incorporating new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. Extensive MATLAB code has been written and is in its final stages of completion, allowing users to analyze complete sets of in-shoe force data and calculate DLS. This revised DLS is being used to formulate the individualized exercise prescriptions used by the exercising subjects. The revised algorithm is currently being implemented into a toolkit for the real-time processing of test subject force-plate data. Details of these findings were recently submitted for publication in the Journal Aerospace and Environmental Medicine. The experiment, which is still underway, should provide a definitive answer to the question of whether or not intermittent load replacement can adequately protect the musculoskeletal system against hypokinetic osteopenia and muscle atrophy and may demonstrate the efficacy of individualized exercise countermeasure planning. As mentioned above, the team is in the process of moving the study to the University of Texas Medical Branch in Galveston, TX for completion of the remaining 12 subjects.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh PR, Licata A, Rice AJ, Gopalakrishnan R, Novotny SC, Kuklis MM, Davis BL, Genc KO, Englehaupt RK, Ilaslan H. "Exercise as a Countermeasure to Bone Demineralization During 12-Weeks of Bedrest." NASA Human Research Program Investigators' Workshop, League City, TX, February 4 – 6, 2008. NASA Human Research Program Investigators' Workshop, League City, TX, February 4 – 6, 2008. , Feb-2008
Articles in Peer-reviewed Journals	Cavanagh PR, Rice AJ, Novotny SC, Genc KO, Englehaupt RK, Owings TM, Comstock B, Cardoso T, Ilaslan H, Smith SM, Licata AA. "Replacement of daily load attenuates but does not prevent changes to the musculoskeletal system during bed rest." Bone Reports. 2016 Dec;5:299-307. http://dx.doi.org/10.1016/j.bonr.2016.10.001 , Dec-2016
Articles in Peer-reviewed Journals	Genc KO, Humphreys BT, Rice AJ, Novotny SC, Englehaupt RK, Gopalakrishnan R, Ilaslan H, Licata AA, Cavanagh PR. "Enhanced Daily Load Stimulus (eDLS): Accounting for Cyclical Loading and Standing on the Osteogenic Response of Bone." Submitted to Aviation Space and Environmental Medicine, July 2008. , Jul-2008